
AWS EKS – Interview Questions & Answers (2 Years DevOps Experience)

1. What is Kubernetes and why is it used?

Kubernetes is a container orchestration platform used to deploy, manage, and scale containerized applications. While Docker helps run individual containers, Kubernetes manages multiple containers across multiple machines. It provides features such as self-healing, load balancing, rolling updates, rollbacks, and automated scaling, which are difficult to manage manually using Docker alone.

2. What is Amazon EKS?

Amazon EKS is a managed Kubernetes service provided by AWS. AWS manages the Kubernetes control plane components like API server, scheduler, and etcd, while users manage the worker nodes. This reduces operational overhead related to cluster setup, upgrades, and security while still giving flexibility to run Kubernetes workloads.

3. What are the main Kubernetes components?

The **control plane** includes:

- API Server – entry point for all Kubernetes operations
- etcd – key-value store for cluster state
- Scheduler – assigns pods to nodes
- Controller Manager – maintains desired state

The **worker node** includes:

- kubelet – communicates with control plane
 - kube-proxy – handles networking
 - container runtime – runs containers
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4. What is a Pod?

A Pod is the smallest deployable unit in Kubernetes. It can contain one or more containers that share the same network namespace and storage. Kubernetes manages Pods, not individual containers.

5. Why does Kubernetes use Pods instead of containers directly?

Pods allow multiple containers to work together as a single unit, such as a main application container and a sidecar container. Pods are ephemeral, so higher-level objects like Deployments are used to ensure availability and scaling.

6. What is a Deployment?

A Deployment is a Kubernetes object used to manage stateless applications. It ensures a desired number of Pods are always running and supports rolling updates and rollbacks.

7. What is a Kubernetes Service?

A Service exposes a set of Pods and provides stable networking. Since Pod IPs change dynamically, Services use labels and selectors to route traffic reliably.

8. Types of Kubernetes Services

- **ClusterIP** – internal-only access
 - **NodePort** – exposes service on node ports
 - **LoadBalancer** – exposes service using cloud load balancer
 - **ExternalName** – maps service to an external DNS name
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9. What is Ingress?

Ingress manages external HTTP/HTTPS access to services inside the cluster. It works with an Ingress Controller (e.g., NGINX or AWS Load Balancer Controller) and supports host-based and path-based routing.

10. What is cert-manager?

cert-manager is a Kubernetes controller that automates TLS certificate creation and renewal using Certificate Authorities like Let's Encrypt. It uses the ACME protocol and creates certificates as Kubernetes Secrets for Ingress.

11. What are PV and PVC?

A **PersistentVolume (PV)** represents storage in the cluster.

A **PersistentVolumeClaim (PVC)** is a request for storage by a Pod.

Pods use PVCs, and Kubernetes binds them to matching PVs.

12. What is a StorageClass?

A StorageClass defines how storage is dynamically provisioned. In EKS, it works with the AWS EBS CSI Driver to automatically create EBS volumes when PVCs are requested.

13. What is IRSA?

IRSA (IAM Roles for Service Accounts) allows Kubernetes service accounts to assume IAM roles. This enables secure, pod-level access to AWS services without using static credentials.

14. How do you troubleshoot a PVC stuck in Pending state?

Check:

1. Pod events
 2. PVC status
 3. StorageClass configuration
 4. EBS CSI Driver pods
 5. IAM permissions (IRSA)
 6. Node availability and AZ alignment
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15. How do you troubleshoot Ingress not working?

Validate in order:

1. Pod health
 2. Service selectors & endpoints
 3. Ingress resource rules
 4. Ingress Controller status
 5. LoadBalancer / DNS
 6. TLS certificate readiness
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16. How do Prometheus, Grafana, and Loki work together?

Prometheus collects metrics, Grafana visualizes them, and Loki stores logs. Promtail runs on nodes to push logs to Loki. Grafana connects to both Prometheus and Loki for unified observability.

17. What is Helm?

Helm is a package manager for Kubernetes. It packages manifests into charts, supports versioning, simplifies deployments, and enables easy rollbacks and upgrades.

18. How do you recover from a failed Helm upgrade?

Check Helm status and history, inspect pod events and logs, fix configuration issues if possible, or rollback to the last stable release using `helm rollback`.

19. Explain Jenkins CI/CD flow for EKS (without ArgoCD)

Jenkins pipeline:

1. Code checkout
2. Build & test
3. Docker image build
4. Push image to ECR
5. Update manifest or Helm values
6. Deploy using kubectl or Helm
7. Validate deployment

20. What is ImagePullBackOff?

ImagePullBackOff occurs when Kubernetes cannot pull an image. Troubleshoot by checking pod events, image name/tag, registry permissions, and imagePullSecrets.

21. ConfigMap vs Secret

ConfigMaps store non-sensitive configuration data. Secrets store sensitive data and are base64 encoded. Secrets should be protected using RBAC and encryption at rest.

22. Why ConfigMap updates don't reflect immediately?

Pods do not automatically reload ConfigMaps. Pods or Deployments must be restarted to pick up updated values.

23. How do you restrict access between namespaces?

Use RBAC to control access to resources and NetworkPolicies to restrict pod-to-pod network traffic.

24. What is a NetworkPolicy?

A NetworkPolicy defines allowed ingress and egress traffic for Pods based on labels, namespaces, and ports. It improves cluster security.

25. Security Groups vs NetworkPolicies in EKS

Security Groups operate at the AWS infrastructure level. NetworkPolicies operate at the Kubernetes pod level.

26. What happens when you update an image in a Deployment?

Kubernetes performs a rolling update by creating new Pods with the updated image while gradually terminating old Pods.

27. How do you rollback a Deployment?

Use:

```
kubectl rollout undo deployment <deployment-name>
```

Kubernetes restores the previous revision.

28. What is CrashLoopBackOff?

CrashLoopBackOff occurs when a container repeatedly crashes. Troubleshoot using pod events, logs, configuration, and resource limits.

29. Why does a node go NotReady?

Common reasons:

- Kubelet failure
- Network issues
- Resource exhaustion
- Node termination

Check node conditions and EC2 instance health.

30. Deployment vs StatefulSet

Deployments are used for stateless applications. StatefulSets are used for stateful applications that require stable identity, persistent storage, and ordered startup.

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